Welfare, Jobs, and Optimal Redistribution: Bridging Discrete Choice and Income Tax Theories*

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1 Objects, observables, and what is random

RURO models explain observed labor supply (or job choice) as the outcome of *(i) preferences* and *(ii) random opportunities*. The central econometric object is the *unconditional* probability (or density) of observing a realized bundle (hours, wage, job attributes), which is shaped jointly by utility and the stochastic availability of market opportunities.

In the classical RURO tradition, market opportunities are indexed by wage–hours pairs (w, h) and a non-market option $(0, 0)$. In our framework the relevant alternative is a *job bundle*

$$j \equiv (w_j, h_j, x_j) \in \mathcal{J},$$

where x_j collects non-pecuniary and institutional characteristics (occupation/task type, contract, amenities, etc.). Let T be the time endowment and $\ell_j = T - h_j$ leisure.

Let I_i denote *non-labor income* (or other exogenous resources). Disposable consumption is

$$C_{ij} \;=\; f(w_j h_j,\, I_i;\, \tau,\, x_j)\,, \tag{1}$$

where $f(\cdot)$ is the net-of-tax mapping induced by tax–benefit rules τ (and possibly job-specific rules through x_j if the tax benefit allows). The key identification role of I_i in several RURO papers is that it shifts consumption while holding (w, h) fixed, and is assumed *not* to shift the opportunity structure (Dagsvik and Jia, 2016).

RURO models use a positive utility index Ψ (often an exponential transform of a deterministic utility) evaluated at the implied consumption–hours bundle:

$$\Psi_i(w, h, x) \;=\; \Psi\Big(C_i(w, h, x),\, h,\, x;\, z_i\Big), \tag{2}$$

where z_i are observed taste shifters. The *random utility* component is introduced through extreme-value/max-stable structures (Fréchet-type constructions) that deliver closed-form choice probabilities (Capéau et al., 2016).

The *random opportunity* component is the (latent) set of market offers available to an individual. In the RURO tradition this is modeled via an offer process (often a Poisson/max-stable construction) that implies an *opportunity density/intensity* over potential offers. Depending on the paper, this density is (i) over hours only, conditional on a wage treated as given; or (ii) over both wages and hours; or (iii) over latent jobs where wages contain an unobserved "ability" component. The econometrician does *not* observe the latent offer set, but observes the realized choice and thus the realized (w, h, x).

2 Canonical RURO likelihood representations

2.1 Hours opportunities with wage treated as given (a common specification)

A standard representation focuses on an opportunity density over hours, $g(h)$, possibly with peaks at standard contracts. Aaberge et al. (2009) show that the probability density of choosing h can be written (suppressing i subscripts) as

$$\phi(h) \;=\; \frac{\exp\Big(v\Big(f(wh, I), h\Big) + (\text{opportunity terms in } h)\Big)}{\int \exp\Big(v\Big(f(wx, I), x\Big) + (\text{opportunity terms in } x)\Big)\; dx}, \tag{3}$$

where the "opportunity terms" parameterize $g(h)$ (including peaks) and the deterministic utility index is $v(\cdot)$ (Aaberge et al., 2009, eq. (14)). This specification is often combined with an estimated/imputed wage equation for non-workers using heckman selection for example; in that sense wages can be treated as *individual-specific* in the empirical implementation even if the choice object is hours (Aaberge et al., 2009).

2.2 Joint wage–hours opportunities: wages as part of the job offer

Capéau et al. (2016) emphasize a job-offer interpretation in which *wages are part of the market opportunity.* Under an independence assumption $g_1(w \mid h) = g_1(w)$, the likelihood to observe a chosen offer (w, h) admits

$$\phi(w,h) = \frac{\Psi(w,h)\ q\, g_1(w)\, g_2(h)}{\Psi(0,0) + \int_{r \in W} \int_{t \in H} \Psi(r,t)\ q\, g_1(r)\, g_2(t)\, dt\, dr}, \tag{4}$$

with the corresponding non-market probability

$$\phi(0,0) = \frac{\Psi(0,0)}{\Psi(0,0) + \int_{r \in W} \int_{t \in H} \Psi(r,t)\ q\, g_1(r)\, g_2(t)\, dt\, dr}. \tag{5}$$

Here q is proportional to the relative arrival intensity of market opportunities (versus non-market), while g_1 and g_2 describe the wage and hours offer distributions (Capéau et al., 2016, eqs. (15)–(15′)).

Capéau et al. (2016) explicitly contrast (4)–(5) with a model where wage is a fixed individual attribute and only hours are chosen from a finite set $\{h_k\}$, yielding

$$\phi(w, h_\ell) = \frac{\Psi(w, h_\ell)}{\Psi(0,0) + \sum_{k=1}^{K} \Psi(w, h_k)}. \tag{6}$$

The difference is that in RURO with wage offers, the denominator aggregates over *all* wage–hours offers, not only over hours conditional on a given wage (Capéau et al., 2016, eq. (16) and discussion). This is exactly why it is not correct to impose a single "wage interpretation" on the whole RURO family: both conventions exist, and the distinction is itself substantive.

3 Identification of preferences versus opportunities

The identification problem is immediate from (4): observed choice frequencies depend on *products* of a preference index and an opportunity density. The literature therefore distinguishes (i) what is *nonparametrically* identified from relative frequencies, and (ii) what requires *parametric* restrictions (functional forms, independence, instruments, normalizations).

3.1 Nonparametric identification

3.1.1 Capéau et al. (2016): identifying offer components from relative proportions

Capéau et al. (2016) (building on earlier work) show that some objects are pinned down by comparing observationally equivalent subgroups and exploiting relative proportions (Capéau et al., 2016, Sec. 3.2).

Fix a group with identical observables (z, I), and consider two subgroups working different hours h_1 and h_2 at the *same* wage w. From (4),

$$\frac{\phi(w, h_1)}{\phi(w, h_2)} = \frac{\Psi(w, h_1) g_2(h_1)}{\Psi(w, h_2) g_2(h_2)}. \tag{7}$$

Varying w identifies the composite object $\Psi(w, h)g_2(h)$ (for $h > 0$), not Ψ and g_2 separately.

Next, compare individuals working the *same* hours h but accepting different wages w_1 and w_2:

$$\frac{\phi(w_1, h)}{\phi(w_2, h)} = \frac{\Psi(w_1, h) g_2(h)}{\Psi(w_2, h) g_2(h)} \cdot \frac{g_1(w_1)}{g_1(w_2)}. \tag{8}$$

Since $\Psi(w, h)g_2(h)$ was identified up to scale from (7), (8) identifies $g_1(\cdot)$ using the normalization $\int_W g_1(w)\, dw = 1$.

Finally, comparing market to non-market choices yields

$$\frac{\phi(w, h)}{\phi(0, 0)} = \frac{\Psi(w, h)\ q g_1(w) g_2(h)}{\Psi(0, 0)}. \tag{9}$$

Normalizing $\Psi(0, 0)$ (a utility normalization) identifies q (Capéau et al., 2016, eqs. (17)–(19)).

What remains nonparametrically unidentified. Even with these steps, *preferences and hours opportunities are not separately identified*: only the product $\Psi(w, h)g_2(h)$ is recovered nonparametrically (Capéau et al., 2016). This is a central limitation.

3.1.2 Dagsvik–Jia (2016): the role of non-labor income and partial identification

Dagsvik and Jia study a closely related latent-job model and make the identification problem explicit. For cross-section data with observed (h, w, I) they obtain (for $h > 0$)

$$\frac{\phi(h, w \mid I)}{\phi(0, 0 \mid I)} = \frac{v\Big(f(hw, I), h\Big)\, \theta\, g_1(h)\, g_2(w \mid h)}{v\Big(f(0, I), 0\Big)}, \tag{10}$$

so that the right-hand side is *nonparametrically identified as a function* of observables (Dagsvik and Jia, 2016, eq. (7) and surrounding discussion). However, it is generally not possible to separately identify $v(C, h)$ and the opportunity term $\theta g_1(h) g_2(w \mid h)$ from (10) alone (Dagsvik and Jia, 2016).

A key exclusion restriction is that *non-labor income enters only through consumption and does not affect the opportunity measure.* This produces variation in C holding (w, h) fixed and yields partial identification results: roughly, $v(C, h)$ is recovered up to an hours-specific multiplicative function,

$$v(C, h) \;=\; \lambda(C, h)\, \delta(h), \tag{11}$$

where $\lambda(C, h)$ is identified while $\delta(h)$ is not (Dagsvik and Jia, 2016, Theorems and identification discussion). Additional restrictions (e.g. independence between offered wages and hours, support and regularity conditions, or parametric forms) are then used to achieve full identification.

3.2 Parametric identification

Because nonparametric separation is limited, most empirical RURO work relies on parametric identification: specify v and specify the opportunity density with a finite-dimensional parameter vector; then estimate jointly by maximum likelihood.

Capéau et al. (2016) adopt Box–Cox-type specifications for preferences and discuss theoretical justifications based on invariance arguments (Capéau et al., 2016). Dagsvik–Jia similarly show that assuming a generalized Box–Cox form for $\log v(C, h)$ plus additional regularity conditions yields full identification (Dagsvik and Jia, 2016, Theorem 4). In this parametric route, the unidentified $\delta(h)$ in (11) is pinned down by functional form and normalization.

Aaberge et al. (2009) parameterize the hours-opportunity density to allow peaks at standard contracts, and estimate these "opportunity" parameters jointly with preference parameters (Aaberge et al., 2009). Capéau et al. (2016) treat q (relative arrival of market versus non-market opportunities) and parameters of $g_2(h)$ (peaks) as opportunity-side objects and highlight the need for normalizations (e.g. utility normalization $\Psi(0,0)$ and opportunity normalizations) (Capéau et al., 2016, Sec. 3.1 and Eq. (24)).

A stronger identification challenge arises when wages contain an unobserved component (ability for eg) that also affects opportunities. Dagsvik–Jia show that identification in such a model requires an exogenous covariate X that affects only the opportunity density (not preferences), entering a wage-offer equation

$$\log W(z) = a + Xb + \eta + \xi(z), \tag{12}$$

with η an individual random effect (ability) and $\xi(z)$ offer-specific noise, and with the opportunity parameter θ allowed to depend on $(a + Xb + \eta)$ (Dagsvik and Jia, 2016, Assumption 7 and eqs. (9)–(10)). we can deduce from This that opportunity-side instruments are not optional rather they are required when unobserved ability loads into wage offers.

4 Our model: identification when jobs are triples (w, h, x)

Classical RURO applications often emphasize (w, h) (possibly adding coarse sectors). Our framework treats a job as a triple (w, h, x) and interprets the opportunity object as *individual-specific access to bundles differentiated by both pecuniary and non-pecuniary characteristics*. Formally, let the (unconditional) probability of observing choice $j = (w, h, x)$ be

$$\phi_i(w, h, x) = \frac{\Psi_i(w, h, x)\ \Omega_i(w, h, x)}{\Psi_i(0, 0, x_0) + \iint\sum_{x\in\mathcal{X}} \Psi_i(r, t, x)\ \Omega_i(r, t, x)\, dt\, dr}, \tag{13}$$

where $\Omega_i(w,h,x)$ is the opportunity density/intensity for offers of type (w,h,x).

Equation (13) makes explicit that choices identify only the composite $\Psi_i \cdot \Omega_i$. Separation therefore requires the same two pillars as in the RURO literature:

1. Exclusion restrictions, Non-labor income I_i and policy variation in the tax-benefit mapping shift C_{ij} in (1) (hence Ψ_i) but are excluded from Ω_i (as in Dagsvik–Jia). Conversely, labor-demand shifters (region×year conditions, occupation-specific unemployment/vacancy indicators) shift Ω_i but are excluded from preferences, conditional on observables.

2. Structure on Ω_i. A possibly workable approach is to factorize the offer intensity,

$$\Omega_i(w,h,x) = q_i(x)\, g_{1i}(w \mid x)\, g_{2i}(h \mid x,w), \tag{14}$$

   with optional simplifications (e.g. conditional independence $g_{2i}(h \mid x,w) = g_{2i}(h \mid x)$) paralleling the independence assumptions used in Capéau et al. (2016).

With (14), Capéau-style relative-frequency arguments extend to identify composite objects such as $\Psi(w,h,x)\, g_2(h \mid x)$ and then recover wage-offer components using wage variation, provided support conditions hold within each x-cell.

5 Unobserved heterogeneity: preferences versus opportunities

RURO models allow unobservables on both sides, but their empirical separability is delicate.

All RURO models include i.i.d. extreme-value/max-stable taste shocks that rationalize probabilistic choice and yield tractable likelihoods. Beyond that, some applications introduce random coefficients (mixed logit) or latent classes for persistent taste heterogeneity; without panel or stated-preference data, these components can be difficult to distinguish from opportunity heterogeneity.

Opportunity heterogeneity is central in the latent-job tradition. Dagsvik–Jia explicitly introduce an individual random effect η in wage offers (12), interpretable as (observed+unobserved) ability loading into opportunities, and require an excluded X for identification (Dagsvik and Jia, 2016). Empirically, Capéau et al. (2016) adopt opportunity shifters such as group-specific unemployment rates in the market-arrival component q to strengthen the opportunity interpretation (Capéau et al., 2016).

References

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